



MOKO SMART



Bluetooth®

Hall sensor

User Guide

MOKO Door Sensor

Version 1.1

Revision History

Version	Date	Notes	Contributor(s)
V1.0	June 25, 2023	Initial version	Daniel
V1.1	Dec 24, 2024	1. Revise the description of hardware reset – “long press button won’t restore the password” in chapter 4.4 – How to restore factory settings .	Daniel

About document

This User Guide was designed to help users to know the overview of **MOKO DOOR SENSOR** product and set up **MOKO DOOR SENSOR** through **MOKO standard Android “MK PIR” APP**. Through this document, users will be initial to know the hardware specifications, installation guidance, application scenarios of Door Sensor and APP user guidance, which including connection, parameters configuration of Beacon advertisement, DFU update and etc.

This guide will not cover the sales administration and the ordering process. Some technical guides will be needed if further explanation is required.

- Applicable version description:

Android APP Version: “MK PIR” V1.0.0 or above

Firmware Version: Door Sensor_V1.0.6 or above

- Each figure (diagram, screenshot, or other image) and table are provided with a number and description:

Figure 1: Typical application scenario of DOOR SENSOR

Table 1: Packing list of DOOR SENSOR

The numbers and descriptions of the figure and table can be found in the “List of Figures” and “List of Tables”.

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1. Overview

The *MOKO Door Sensor*, integrated with data logging and real-time status alerts, is a BLE 4.2 based wireless Beacon to detect when doors open or close for powerful visibility.

With precise sensors and an easy-to-install design, the door monitor provides powerful visibility and improved security to both roll-up and swing (i.e., hinge) doors without any complicated wiring or cables. It also delivers enhanced visibility of trailer and container operations and improved cargo security, which enables fleet operators to know if the cargo area is being accessed as scheduled and react quickly to unexpected openings.

Thanks to its outstanding 2600mAh large battery, the door monitor could run for 8+ years to eliminate the maintenance costs.

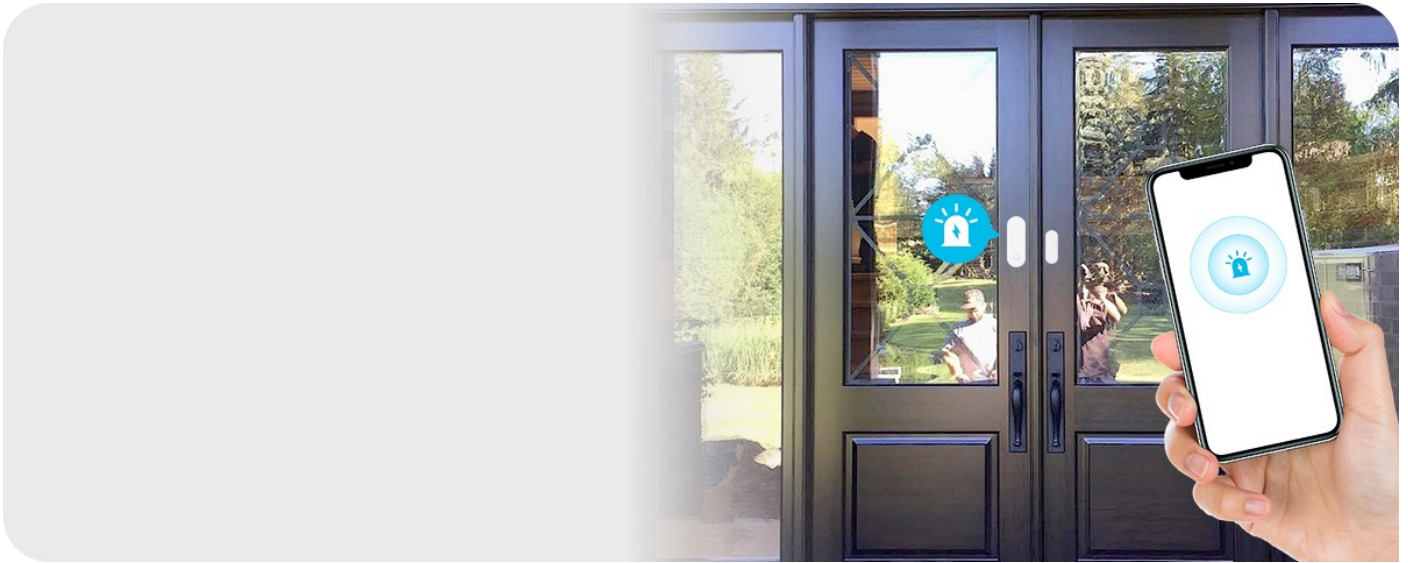


Figure 1: Typical application scenario of DOOR SENSOR

2. Getting to know DOOR SENSOR

2.1 What's in the box

You will find the following items per DOOR SENSOR unit in the packing box:

Item	Quantity	Remark
DOOR SENSOR	1	Main product
Bracket holder	1	Wall-mounted holder
Magnet	1	Magnet accessory

Table 1: Packing list of DOOR SENSOR

Remark: Packing method can be customized vary from different orders.

2.2 Appearance

2.2.1 Dimensions

Here we have described the dimensions of DOOR SENSOR as below:

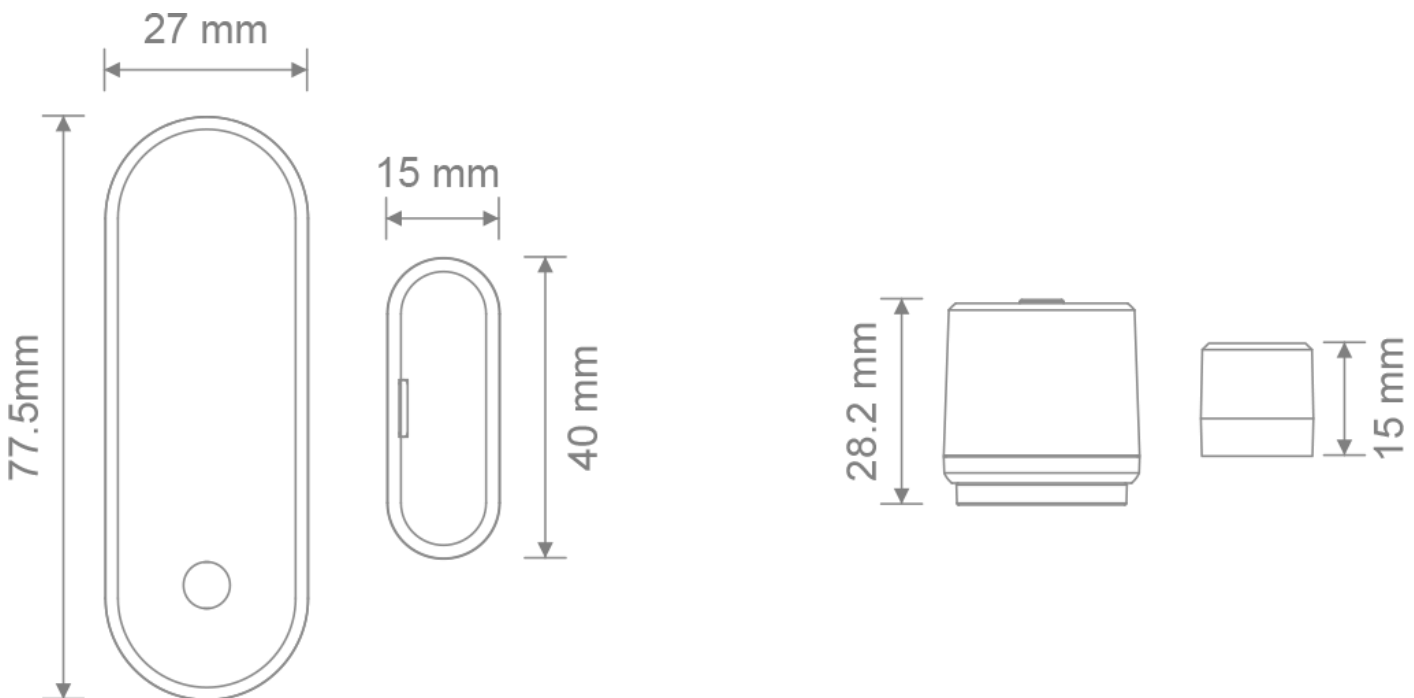
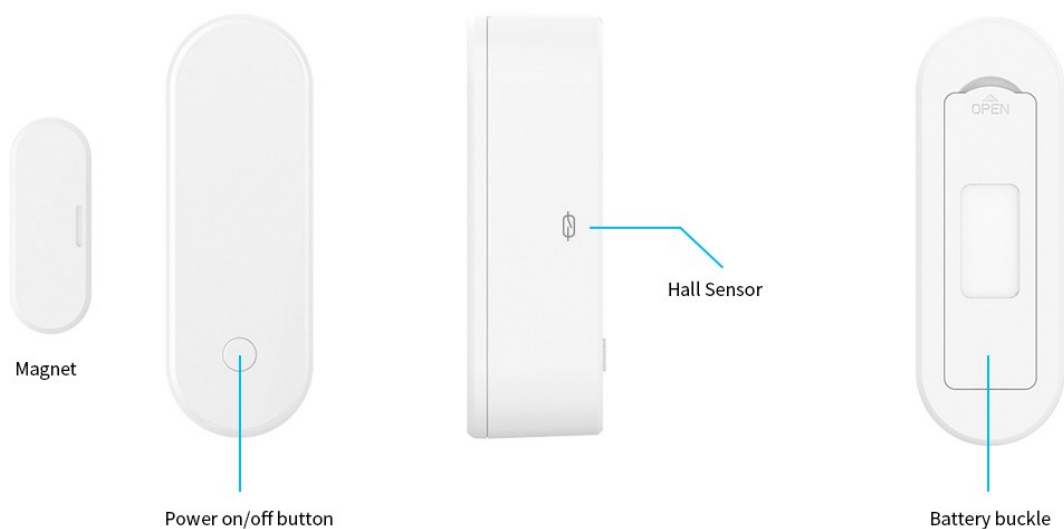


Figure 2: Dimensions of DOOR SENSOR

2.2.2 Function identifications

The DOOR SENSOR product's function identifications are shown in the figure below:



- **Power On/Off button** – Push the button(long press for 3s) to turn on or turn off the product.
- **Hall sensor** – Magnetic detection(Door/Window on/off detection) part.
- **Battery buckle** – Open battery room and replace battery through this buckle.

2.3 Specifications

2.3.1 General specifications

General specifications	
Main chip	Nordic nRF52 series
Wireless connectivity	Bluetooth 4.2(Hardware compatible with Bluetooth 5.1)
Dimensions	77.5* 27.0*24.7mm(Body with bracket holder) 40.0*15.0*15.0mm(Magnet)
Weight	30g
Material	PC
Advertising range	Up to 150 meters (in the open area and no obstacles)
IP rating	IPX4
Color	White
Installation	Screw, Sticker
Button	Power On/Off button
LED	Red LED
Built-in sensor	Accelerometer sensor(Optional) Hall sensor
Operating temperature	-20°C / + 60°C
Storage temperature	-20°C / + 60°C (without battery) 10°C / + 25°C (with battery)
Storage humidity	0% ~ 95% (non-condensing)
Antenna type	PCB on-board antenna

General specifications	
Power supply	Replaceable 1*AA battery 2600mAh capacity
Certification	CE, FCC and RoHS, REACH compliant

Table 2: General specifications of DOOR SENSOR

2.3.2 Hall Sensor Specifications

Hall sensor specifications	
Detection distance	20mm-25mm
Recommended installation gap	< 10mm
Status indication	Magnet away (Door open): 1 Magnet approaching (Door closed): 0

Table 3: Specifications of Hall sensor

2.3.3 Battery consumption

Battery consumption				
Accelerometer sensor	Adv interval	Tx Power	Consumption	Estimated lifespan
Not equipped	1000ms	0dBm	18.5uA	8+ years

Table 4: Battery consumption of DOOR SENSOR

2.3.4 LED descriptions

LED response status	
Power ON	Blinking for 2 seconds
Power OFF	Solid for 2 seconds
Hardware/Software reset	Blinking for 2 seconds and then device reboot
Low battery	Blinking 2 times every 10 seconds
Device connect	Blinking for 2 seconds
DFU upgrade	Blinking during DFU and blinking for 2 seconds to reboot

Table 5: LED response status in various situations

3. How the *DOOR SENSOR* works?

The *DOOR SENSOR* is based on hall sensor, to detects hall sensor status respect to magnet (i.e., door closed/open). The typical logical is to detects the hall sensor status, and then upload the status to the gateway node through BLE advertisement. So, we will describe door open/close status detection and BLE advertisement separately.

3.1 Door open/close status detection

In *DOOR SENSOR*, we have integrated with the hall sensor (need to be applied together with magnet accessory) to monitor the door's open/close status.

You can refer to "[chapter 4.1 – How to install DOOR SENSOR](#)" to install the *DOOR SENSOR* as well as magnet accessory. When magnet is approaching to hall sensor and distance is less than 20mm, hall sensor will output high level and device will report "1" in the BLE advertisement to the gateway for processing and delivery to cloud services.



Figure 3: How to check door's status in the advertisement?

3.2 Bluetooth advertisement

In last chapter, we have mentioned that the DOOR SENSOR is uploading the sensor status or other basic information through BLE advertisement, and that basic information can also be configured through MOKO "MK PIR" APP. Before introducing the APP user manual, we will guide you to fully understand the advertisement format and content parsing rules.

Regarding of PIR series firmware, there have only one customized advertisement format, which is a combination of *ADV_IND* (Advertising packet) and *SCAN_RSP* (Scan response packet). In the next we will describe the detailed information of these two packets.

- **ADV_IND (advertising packet)** ---- Mainly to express status of hall sensor, battery voltage, Major, Minor and etc.
- **SCAN_RSP (scan response packet)** ---- Mainly to express Tx power and device name.

Below picture showed the example content of Beacon advertisement:

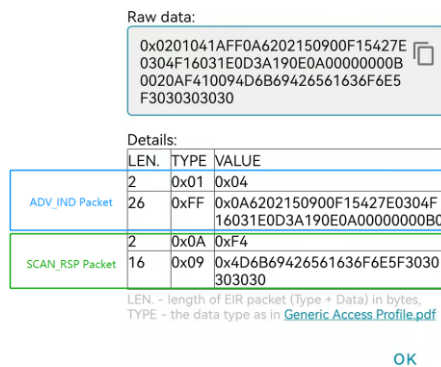


Figure 4: Example content of Beacon advertisement

Advertising Format - ADV_IND packet				
Example	Parameters		Descriptions	Length
0x02	ADV Length		ADV length of "Flags" content	1 byte
0x01	ADV Type		ADV type: Flags	1 byte
0x04	Beacon Type		BR/EDR not supported	1 byte
0x1A	ADV Length		ADV length of "Manufacturer data" content	1 byte
0xFF	ADV Type		ADV type: Manufacturer data	1 byte
0x0A 62	Company ID		Value = 0x0A 62(MOKO TECHNOLOGY Ltd)	2 bytes
0x02	iBeacon Type		Value = 0x02 (Proximity beacon, fixed value)	1 byte
0x15	iBeacon Length		Value = 0x15 (21 bytes)	1 byte
0x09	Device status	RFU	Occupy 2 bits (bit 6 - bit 7), Reserved for future use	1 byte(2 bits)
		PIR delay response status	Occupy 2 bits (bit 4- bit5). 10: High delay ; 01: Medium delay ; 00: Low delay	1 byte(2 bits)
		Door Open/Close status	Occupy 1 bit (bit 3). 0: Close (Door magnetic approaching); 1: Open (Door magnetic away)	1 byte(1 bits)
		PIR sensor sensitivity	Occupy 2 bits (bit 1- bit2). 10: High sensitivity ; 01: Medium sensitivity ; 00: Low sensitivity	1 byte(2 bits)
		PIR sensor detection status	Occupy 1 bit (bit 0). 0: No effective motion detected ; 1: Effective motion detected	1 byte(1 bit)
0x00	RFU		1 byte reserved for future use , default 0x00	1 byte
0xF1 54 27 E0 30 4F	MAC address		6 bytes MAC address, read from chipset	6 bytes
0x16 03 1E 0D 3A 19	Timestamp		Timestamp format : YY/MM/DD/HH:MM:SS (Content parse: 21/03/30/13:58:25)	6 bytes
0x0E 2A	Battery voltage		Unit: mV. (HEX to DEC) (Content parse: 3626mV)	2 bytes
0x00 00	Major		Can be customized as dividing device group	2 bytes
0x00 00	Minor		Can be customized as device ID	2 bytes
0xB0	RSSI@1m		Calibrated RSSI at 1m, Range: -100dBm ~ 0dBm (Content parse: -80dBm)	1 byte

Table 6: Content structure of Customized-PIR advertising packet

Advertising Format - SCAN_RSP packet				
Example	Parameters		Descriptions	Length
0x02	ADV Length		ADV length of "Tx Power" content	1 byte
0x0A	ADV Type		ADV type: Tx Power	1 byte
0xF4	Tx Power		1 byte signed integer, unit: dBm. (Content parse: -12dBm)	1 byte
0x10	ADV Length		ADV length of "Complete Local Name" content	1 byte
0x09	ADV Type		ADV type: Complete Local Name	1 byte
0x4D 6B 69 42 65 61 63 6F 6E 5F 30 30 30 30 30	Complete Local name	Device name	1-10 bytes characters, comply with US-ASCII standard. (Content parse: MklBeacon_)	1-10 bytes
		Serial ID	1-5 bytes decimal numeral, Configuration range: 0 - 99999 (Content parse: 00000)	1-5 bytes

Table 7: Content structure of Customized-Device info response packet

4. Basic instructions

4.1 How to install *DOOR SENSOR* ?



Figure 5: How to install *DOOR SENSOR*?

4.2 How to power ON/OFF *DOOR SENSOR* ?

Regarding of *DOOR SENSOR*, you can power on/off it through the mechanical button with below operations:

- **Power ON:** Long press button for more than 3 seconds, and red LED will keep blinking for 2 seconds.
- **Power OFF:** Long press button for more than 3 seconds, and red LED will keep solid for 2 seconds.

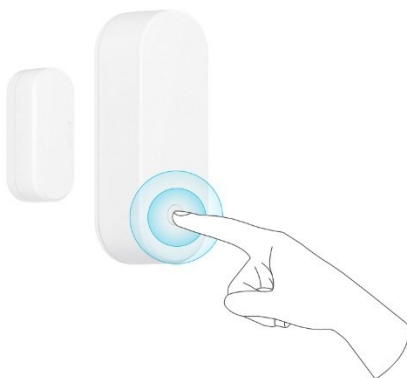


Figure 6: How to power on/off *DOOR SENSOR*?

4.3 How to replace battery ?

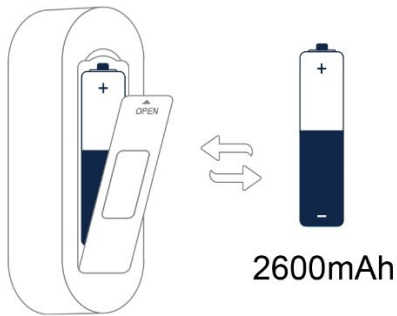


Figure 7: How to replace battery in DOOR SENSOR?

4.4 How to restore factory settings ?

There have two ways to restore factory settings.

- **“MK PIR” APP (Software reset):** Connect DOOR SENSOR with “MK PIR” APP and then execute “Remote reset” operations to finish the software reset.
- **Power Button (Software reset):** In power off status, long press button for more than 10s (during this period, LED will blink to indicate the device ON, please ignore and keep long press) and then release when you see the LED solid lightening, to finish the software reset.

Remark: Software reset will not reset connection password.

5. MOKO PIR APP user guidance

This user guidance for “MK PIR” APP is based on version 1.0.3 or above (only android application available now), and the contents of this user guidance are subject to change without prior notice for further improvement. For latest APP, please download from Google play store.

Remark: MOKO Door Sensor is working with “MK PIR” series firmware, so it is compatible with “MK PIR” APP. And you can ignore other sensor information in the APP and only focus on the Door Sensor information regarding of Door Sensor product.

5.1 Discover your Door Sensor

The smartphone with “MK PIR” APP installed can be as a *Bluetooth Center* to scan and discover the advertisement packets of Bluetooth Low Energy (BLE) Beacons, and it will provide multiple strategies to filter the Bluetooth packets and that can help you to capture the required BLE peripherals directly and quickly.

5.1.1 Scanning your Door Sensor

When entering into the “**SCANNING PREVIEW**” interface, the *SCANNER* will start scanning and list all surrounding advertising Bluetooth LE peripherals which integrated with MOKO PIR series firmware, and you can stop scanning manually by pressing scanning button “↺”.

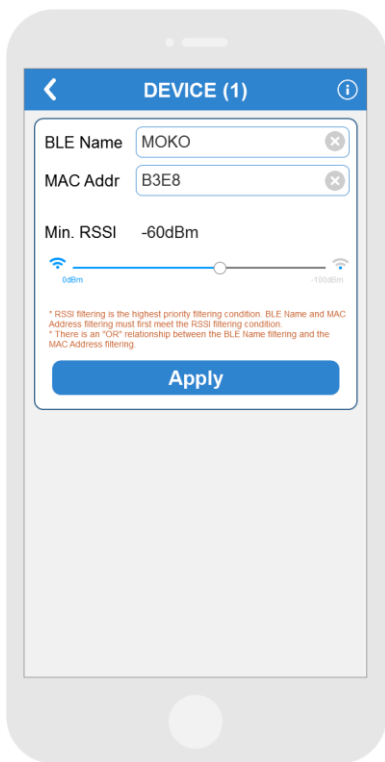
In this *SCANNER* interface, you will see device advertisement preview to recognize your Beacon device. To distinguish specific type of Beacons separately, only Door Sensor can be shown in the APP preview interface and others will be filtered automatically. Below picture shows the default scanning preview of *MOKO DOOR SENSOR* device.



Figure 8: Default scanning preview of DOOR SENSOR

5.1.2 Filtering your Door Sensor

To find your Door Sensor quickly, we have provided multiple strategies to filter among mass Beacon devices, users may narrow your device list by BLE name, MAC address and RSSI value through “**Edit Filter**” option.



BLE Name	Allow to input 1-10 characters. Example: Only Door Sensors whose BLE name contain “MOKO” will be displayed in SCANNER interface. NOTE: Space effective, and case insensitive as well.
Mac Addr	Allow to input 1-6 bytes (HEX). Example: Only Door Sensors whose Mac address contain “B3:E8” will be displayed in SCANNER interface. NOTE: There is no need to input colon between MAC address in search filter column, and case insensitive as well.
Min.RSSI	Only the advertisement packets with RSSI not less than the Min.RSSI value will be displayed in SCANNER interface , and configuration range is -100dBm to 0dBm. Example: When setting the Min.RSSI value to -70, the ADV packets with RSSI of -70dBm or greater than -70dBm will be list in SCANNER interface.

Figure 9: Configurations of scanning filter settings

NOTE:

- ① RSSI is the highest priority filtering condition. It means that filtered device must meet the RSSI filtering condition.
- ② There is an “OR” relationship between BLE Name and Mac Addr. It means that filtered device just need meet one of the filtering conditions.

5.2 Connect with your Door Sensor

For Door Sensors, if the status is connectable, then it will shows “**CONNECT**” button in the “**SCANNING PREVIEW**” interface. You can follow below steps to connect with your Beacon device:

Step 1: Press “**CONNECT**” button, red LED will blink 2seconds rapidly.

Step 2: Input connection password and then press “**OK**” button, red LED will blink 2seconds rapidly again.



Notice – Default password in MOKO standard firmware is “Moko4321”.

5.3 Getting your Door Sensor to broadcast

Once device connected and the main screen of the “MK PIR” APP appears, including **ADVERTISEMENT interface**, **SETTING interface** and **DEVICE interface**.

Remark: If there are no write/read operations within 2 minutes after device connected, then it will be disconnected automatically to save power consumption.

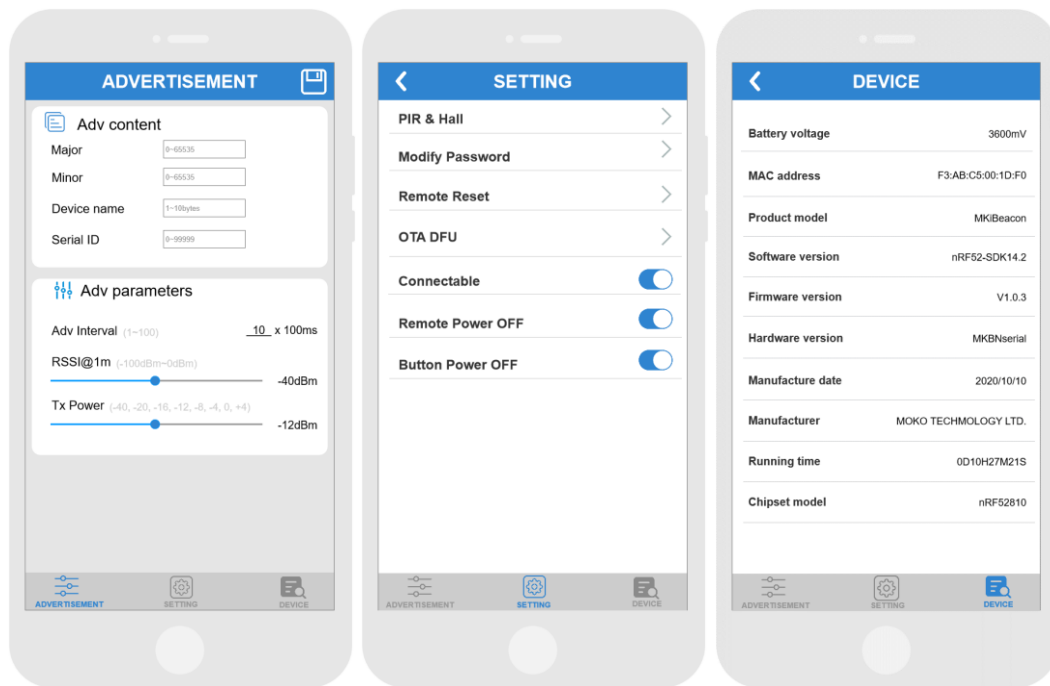


Figure 10: Configuration interface overview of main screen in “MK PIR” APP

5.3.1 Setting up advertisement content

After you connected with device, then you can go to “**ADVERTISEMENT >>> Adv content**” part to configure the corresponding advertisement content. Details are as below.

Adv content

Major

0~65535

Minor

0~65535

Device name

1~10bytes

Serial ID

0~99999

Figure 11: How to configure advertising content?

- **Major** ---- You can define the type of value as you want in practical scenario, such as dividing region, group or etc. Configuration range: 0-65535
- **Minor** ---- You can define the type of value as you want in practical scenario, such as specific room, device or etc. Configuration range: 0-65535
- **Device name** ---- ASCII characters within 10 bytes.
- **Serial ID** ---- Serial ID from 0 to 99999.

Remark: Complete local name is a combination of “Device name” and “Serial ID”. For instance, if you have set “PIR” as device name, and “00001” as serial ID, then it will show “PIR00001” in the preview interface. Or if you checked with “nRF Connect” APP, the complete local name will show “PIR00001”.

5.3.2 Setting up advertising parameters

After you connected with device, you can also go to “**ADVERTISEMENT >>> Adv parameters**” part to configure the corresponding advertisement parameters. Details are as below.

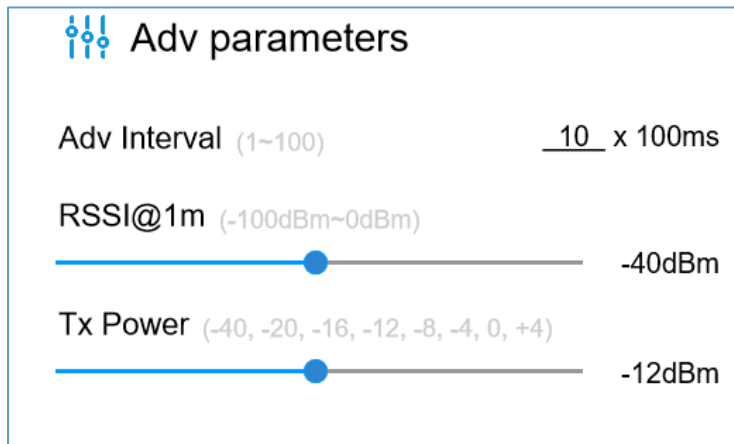


Figure 12: How to configure advertising parameters?

- **Adv Interval** ---- The time between advertising packets. Configuration range: 100ms – 10000ms.
- **RSSI@1m** ---- Value that declares to receiving devices what the power should be at 1m distance from the Beacon. Configuration range: -100dBm – 0dBm.
- **Tx Power** ---- Output power in number of dBm. Configuration level: -40/-20/-16/-12/-8/-4/0/+4 dBm.

5.4 Access to Hall Sensor status

Currently we have adopted “door/window intrusion alarm” functions for “MOKO Door Sensor” products. For the door sensor status monitoring, you can go to “**SETTING >>> PIR & Hall**” interface for further configurations.

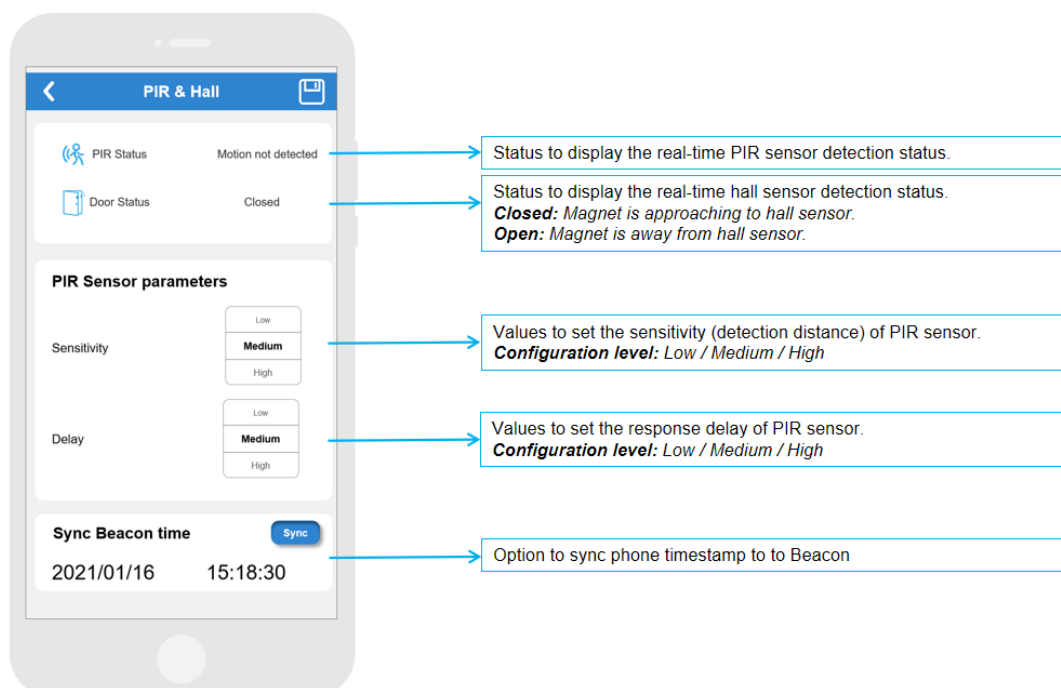


Figure 13: Configurations of Hall sensor

5.5 Modify connection password

In *DOOR SENSOR* firmware, “Moko4321” was set as default connection password. You can try to modify your Beacon’s connection password by just tapping “**SETTING >>> modify password**” option, and then input the password as you want.

Remark: Connection password must be 8 characters in length.

5.6 Switch your Door Sensor to unconnectable

In some cases, you will need to disable the Beacon connectable property, to avoid high power consumption caused by deliberate connections. So, you can disable the connectable status in “**SETTING >>> Connectable**” option. After disabled, device will not be connected by master BLE device anymore unless reset the Beacon.



Figure 14: Switch your Door Sensor to unconnectable status

5.7 Disable button power off function

Normally the button is used for power on/off purpose, however in some specific cases, there will need to disable the button function in order to avoid deliberate operations, so logically there also should have the corresponding switch to enable/disable it. You can go to “**SETTING >>> Button Power OFF**” to disable the button power off functions.

- **Button power off** ----- You will not be able to turn off the device through mechanical button once disabled.



Figure 15: Disable “button power off” function

5.8 Remote turn off your Door Sensor

In some cases that mechanical button function should be disabled to avoid intentional shut down situations, then we will need to remote shut down function for backup.

Under this circumstance, you can remote turn off your Beacon by just tapping “**DEVICE >>> Remote Power OFF**” option, then it will give prompt to you to confirm the shut down operations.



Figure 16: Remote power off your Door Sensor



Notice – Please make sure that device have mechanical button, or other ways to turn on before you do remote shut down operations.

5.9 Remote reset your Door Sensor

In some cases that mechanical button function should be disabled or without mechanical button, to avoid intentional reset situations, then we will need remote reset function for backup. You can remote reset your Beacon by just tapping **"DEVICE >>> Remote reset"** option, then it will give prompt to you to confirm the reset operations.

Remark: Remote reset operations will not change the password to factory reset status. If you want to totally reset the Beacon with password restore as well, then you may need to do hardware reset operations.

5.10 Check your device information

In some cases, you will need to know the firmware version of Beacon, or product model and etc. You can access to **"Device"** interface to check and confirm that essential information, which including battery voltage, running time since power on, MAC address and etc.

5.11 Upgrade your device's firmware

The Beacon can realize OTA firmware update through DFU operations. In the next we will describe the detailed steps based on android **"MK PIR"** APP.

Step 1: Copy the upgrade package to a folder in the root directory of the smartphone.

Step 2: **Connect** with device and go to **"SETTING"** interface, press **"OTA DFU"** option and then file manager route of smartphone will be prompted.

Step 3: Choose the required upgrade firmware package and the upgrade process will start automatically. Wait for the upgrade to complete, APP will disconnect with Beacon after upgrade completed successfully.

Remark: Please contact with our sales team for latest firmware OTA package if need.

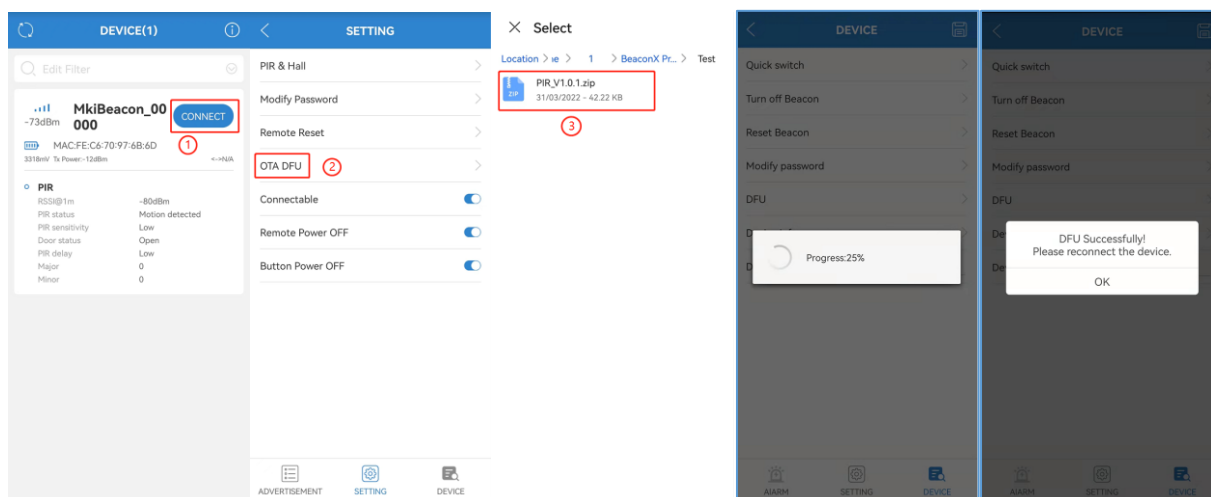


Figure 17: Firmware upgrade workflow of "MK PIR" APP (Android)

5.12 Obtain log file from Android APP

Sometimes when testing with our APP, there may occurred some issues that cannot be reproduced in MOKO side, then we will need you to help obtain the log file from android APP and then provide for us to further analysis.

Currently only Android APP support log record, and the **LOG** option is in “**About**” interface, you can export the log file and then send to us by following below steps if rare issue happened.

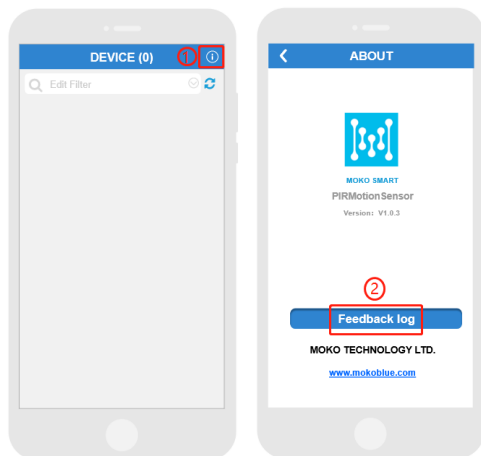


Figure 18: Obtain log file from “MK PIR” APP

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